Congenital Plasminogen Activator Inhibitor Type 1 (PAI-1) Deficiency — Comprehensive Disease Characterization

Causal gene: SERPINE1 (encoding plasminogen activator inhibitor-1, PAI-1). **MONDO:0013227 · OMIM:613329.**

> **Summary answer.** Congenital PAI-1 deficiency is an extremely rare, autosomal-recessive bleeding disorder caused by biallelic loss-of-function mutations in **SERPINE1**, the gene that encodes plasminogen activator inhibitor-1 (PAI-1), the principal physiologic inhibitor of tissue-type (tPA) and urokinase-type (uPA) plasminogen activators. Absence of PAI-1 removes the brake on plasminogen activation, causing a hyperfibrinolytic state in which hemostatic fibrin clots are degraded prematurely. Clinically this produces moderate, characteristically **trauma-/surgery-provoked and delayed bleeding** (plus menorrhagia and easy bruising) with **normal routine coagulation tests**; it responds well to antifibrinolytic drugs (tranexamic acid, ε-aminocaproic acid). Chronic complete deficiency also has extra-hemostatic consequences (notably cardiac fibrosis) and pleiotropic effects on cellular senescence/aging.

Evidence base: predominantly **human clinical** (case reports and the Old Order Amish founder kindred), supported by **model-organism** (Serpine1-knockout mouse), **in vitro** (CRISPR knockout endothelial cells and iPSC-cardiomyocytes), and **computational/structural** data. There is no large registry; most knowledge is disease-level/aggregated from small series plus one extended pedigree, not population EHR.

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1. Disease Information

Overview. PAI-1 deficiency is a disorder of the fibrinolytic (clot-dissolving) system. PAI-1 normally down-regulates fibrinolysis; when it is absent, plasminogen activators act unopposed, plasmin is generated in excess, and fibrin clots lyse prematurely, producing a

bleeding diathesis (Jain & Acharya 2019, ; PMID 31427261 Mehta & Shapiro 2008,). PMID 19141166

Key identifiers. - **MONDO:** MONDO:0013227 - **OMIM:** 613329 (Plasminogen activator inhibitor-1 deficiency) - **Gene:** SERPINE1 (HGNC:8583; NCBI Gene 5054; Ensembl ENSG00000106366; UniProt protein P05121), chromosome **7q22.1** - **Orphanet:** ORPHA:465 (congenital PAI-1 deficiency) - **ICD-10:** D68.8 (other specified coagulation defects) — no PAI-1-specific code; **ICD-11:** 3B19.Y (other specified coagulation defects) / fibrinolysis defect - **MeSH:** no dedicated descriptor; indexed under *Plasminogen Activator Inhibitor 1* (D017395) and *Hemorrhagic Disorders*

Synonyms / alternative names. Congenital PAI-1 deficiency; complete PAI-1 deficiency; plasminogen activator inhibitor type 1 deficiency; PAI-1 deficiency; SERPINE1-related bleeding disorder.

Data source type. Disease-level aggregated knowledge drawn from individual patient reports and one large, well-characterized pedigree (Old Order Amish). No population-scale EHR or registry prevalence exists; the Dutch RBiN cohort (PMID 42780071) is among the few systematic patient series (14 PAI-1-deficient patients).

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2. Etiology

Causal factor — genetic. The disease is monogenic: homozygous or compound-heterozygous loss-of-function variants in **SERPINE1** abolish functional PAI-1. The canonical mutation is the Old Order Amish founder frameshift **SERPINE1 c.699_700dupTA** (exon 4), causing complete, "absolute" PAI-1 deficiency (Fay et al. 1997, ; PMID 9207454 Khan et al. 2021,). PMID 33439236

It is **not** infectious or primarily environmental.

Genetic risk factors. Biallelic null SERPINE1 genotype is both necessary and sufficient. Carrier (heterozygous) status is clinically silent for bleeding (Fay 1997). Consanguinity/endogamy (as in the Amish) raises homozygosity risk; a founder effect concentrates the c.699_700dupTA allele in that community.

Environmental risk factors (for bleeding events, not for the disease itself). Trauma, surgery, dental extraction, menstruation, childbirth, and antithrombotic/antiplatelet drugs unmask or worsen bleeding. Bleeding is overwhelmingly **provoked** rather than spontaneous (Mehta & Shapiro 2008).

Protective factors. Antifibrinolytic prophylaxis (tranexamic acid, ε-aminocaproic acid) prevents provoked bleeding. Intriguingly, at the systemic level **loss of PAI-1 is itself "protective" against aging/metabolic pathology**: PAI-1 is a pro-senescence factor, and null carriers show longer leukocyte telomeres, lower fasting insulin, and extended lifespan (Khan et al. 2017, P 29152572) — a protective-direction pleiotropy opposite to the hemostatic deficit.

Gene–environment interaction. Genotype sets a lifelong hyperfibrinolytic baseline; the *phenotype* (a bleed) emerges only when a hemostatic challenge (trauma/surgery/menses) occurs. In mouse models, hemodynamic stress (angiotensin II, norepinephrine) is required to reveal cardiac hemorrhage/fibrosis in PAI-1-/- animals (Pettey et al. 2025, P 41332700), a clear G×E pattern.

3. Phenotypes

All manifestations are hemostatic or their sequelae. Onset is **congenital** (lifelong), severity **mild-to-moderate**, course **episodic/provoked and non-progressive** (Mehta & Shapiro 2008, P 19141166; Fay 1997, P 9207454).

Phenotype	Type	Characteristics / frequency	Suggested HPO
Abnormal bleeding tendency	Clinical sign	Core feature, provoked/ delayed; essentially all homozygotes	HP:0001892 (Abnormal bleeding)
Prolonged/delayed bleeding after surgery or trauma	Clinical sign	Hallmark; "delayed rebleeding" after initial hemostasis	HP:0004846 (Prolonged bleeding after surgery)
Menorrhagia / heavy menstrual bleeding	Symptom	Common and often most burdensome in affected women	HP:0000132 (Menorrhagia)
Epistaxis	Symptom	Reported, mild-moderate	HP:0000421 (Epistaxis)
Easy bruising	Clinical sign	Frequent	HP:0000978 (Bruising susceptibility)
Joint bleeding (hemarthrosis)	Clinical sign	After mild trauma; reported in severe cases	HP:0005261 (Hemarthrosis)
Intracranial hemorrhage	Clinical sign	After mild head trauma; serious but uncommon	HP:0002170 (Intracranial hemorrhage)
Postpartum/obstetric hemorrhage	Clinical sign	Risk period in affected women	HP:0011891 (Postpartum hemorrhage)
Myocardial/cardiac fibrosis	Physical/ imaging	6/9 (67%) young adult homozygotes with late gadolinium enhancement vs 0/8 controls (P=0.001)	HP:0001637 (Abnormal myocardium/ fibrosis)
Enhanced fibrinolysis / shortened clot lysis	Laboratory abnormality	Universal; functional diagnostic signature	HP:0040239 / lab finding
Delayed wound healing	Clinical sign	Reported with congenital deficiency (in vitro corroboration)	HP:0001058 (Poor wound healing)

Quality-of-life impact. Generally modest between events; episodic bleeds (especially menorrhagia, surgical/obstetric bleeding) can cause anemia, activity restriction, and procedural anxiety. With antifibrinolytic management, daily functioning is typically near-normal. No validated disease-specific QoL instrument exists; generic bleeding-disorder tools (e.g., ISTH-BAT bleeding assessment) apply.

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4. Genetic / Molecular Information

- **Causal gene: SERPINE1** (HGNC:8583; OMIM gene 173360; 7q22.1), encoding PAI-1 (serpin clade E, member 1).
- **Pathogenic variants.** Reported variants are **loss-of-function/null**:
- **c.699_700dupTA** (exon 4 frameshift) — Amish founder allele; **germline**, classified **pathogenic** (ACMG: PVS1 null + segregation + absent antigen) (Khan 2021, P 33439236; Fay 1997, P 9207454).
- Other families carry distinct frameshift/truncating SERPINE1 defects (only a few genetically characterized; Mehta & Shapiro 2008, P 19141166).
- **Variant class / consequence:** frameshift → premature stop → **complete loss of function** (no measurable PAI-1 antigen/activity). Not gain-of-function, not dominant-negative.
- **Allele frequency:** biallelic null genotype is vanishingly rare in gnomAD; the founder allele is enriched only in the Amish subpopulation.
- **Somatic vs germline:** germline.
- **Modifier genes:** other fibrinolytic components (plasminogen, α2-antiplasmin, TAFI, tPA/uPA levels) and the common **SERPINE1 promoter 4G/5G polymorphism (rs1799889)** modulate *PAI-1 level* in the general population but are not the cause of congenital deficiency; they are plausible phenotype modifiers.
- **Epigenetic information:** SERPINE1 expression is strongly inducible (TGF-β, hypoxia, inflammation) but no disease-specific epigenetic lesion is implicated in congenital deficiency.

- **Chromosomal abnormalities:** none; this is a single-gene point/indel disorder.

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5. Environmental Information

- **Environmental factors:** none cause the disorder. Environmental/physiologic *triggers of bleeding* include trauma, surgery, dental procedures, menstruation, and childbirth. Antithrombotics/NSAIDs can aggravate bleeding.
- **Lifestyle factors:** contact sports and activities with trauma risk increase bleed likelihood; alcohol may affect fibrinolysis marginally.
- **Infectious agents:** not applicable — non-infectious Mendelian disease.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic null **SERPINE1** mutation (e.g., c.699_700dupTA) **leads to** absence of functional PAI-1 protein (loss of function; confirmed by undetectable PAI-1 antigen/activity).
2. Loss of PAI-1 **results in** failure to inhibit tissue-type (tPA) and urokinase-type (uPA) plasminogen activators (de-repression of the principal serpin brake).
3. Unopposed tPA/uPA activity **leads to** increased conversion of plasminogen to **plasmin** (enhanced plasmin generation — demonstrated in patients and in CRISPR PAI-1-knockout endothelial cells).
4. Excess pericellular and fluid-phase plasmin **results in** accelerated proteolysis of fibrin (and fibrinogen) — i.e., **hyperfibrinolysis**.
5. Hyperfibrinolysis **leads to** premature dissolution of newly formed hemostatic clots → **delayed rebleeding** after an initially normal clot forms.
6. When a hemostatic challenge (trauma, surgery, menses, delivery) occurs, premature clot lysis **results in** the clinical bleed (mucosal, surgical, joint, intracranial). *(Branch A — the bleeding phenotype.)*
7. **Branch B (tissue remodeling, chronic):** chronically unopposed plasmin/plasminogen activation **leads to** excess activation of matrix metalloproteinases and degradation of vascular/extracellular matrix; under hemodynamic stress this **results in** microvascular

rupture, hemorrhage, and subsequent reparative/replacement **cardiac fibrosis** (demonstrated in PAI-1-/- mice; associated with late gadolinium enhancement in human homozygotes — the human step is *associative/inferred*, mouse step *demonstrated*).

8. **Branch C (senescence/aging):** because PAI-1 is a pro-senescence SASP effector downstream of TGF-β, its absence **leads to** reduced cellular-senescence signaling, associated with longer telomeres and favorable metabolic/longevity measures (*inferred* from carrier associations and murine accelerated-aging models).

Supporting detail by category. - *Molecular pathways:* fibrinolysis / plasminogen-activation system (KEGG "Complement and coagulation cascades" hsa04610; Reactome "Dissolution of Fibrin Clot" R-HSA-75205). PAI-1 is a downstream effector of **TGF-β** signaling (relevant to fibrosis) and of hypoxia/inflammatory induction. - *Cellular processes:* pericellular proteolysis, cell migration/adhesion (PAI-1–vitronectin axis), angiogenesis, inflammation, and **cellular senescence** (GO:0090398). PAI-1-KO endothelial cells show disordered, multidirectional migration and impaired directed wound closure (Sano et al. 2026,).

P 42946217

- *Protein dysfunction:* **loss of a metastable serpin.** Active PAI-1 is metastable and normally converts to a latent form; it inhibits proteases via its reactive-center loop and binds vitronectin (Sharp et al. 1999,).

P 10368279

In disease the protein is simply **absent** (truncated), not misfolded/aggregated. - *Tissue-damage mechanisms:* hemorrhage, and (under stress) plasmin-mediated matrix degradation → fibrosis. - *Biochemical abnormality:* absent serine-protease-inhibitor (anti-tPA/anti-uPA) activity. - *GO terms:* serine-type endopeptidase inhibitor activity (GO:0004867); negative regulation of fibrinolysis (GO:0051918); plasminogen activation (GO:0031639); negative regulation of plasminogen activation. *CL cell types:* endothelial cell (CL:0000115), hepatocyte (CL:0000182, systemic PAI-1 source), platelet (CL:0000233, α-granule PAI-1 store), cardiomyocyte (CL:0000746), adipocyte (CL:0000136).

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7. Anatomical Structures Affected

- **Organ/system level — primary:** blood & hemostatic system (UBERON:0000178 blood; cardiovascular system UBERON:0004535). Bleeding sites: nasal mucosa (epistaxis), oral

cavity/gingiva, uterus/endometrium (menorrhagia), joints (hemarthrosis), and brain (intracranial hemorrhage).

- **Secondary/complication organs: heart** (myocardial fibrosis; UBERON:0000948), **brain** (UBERON:0000955, hemorrhage), musculoskeletal (joints UBERON:0001485), and **skin/soft tissue** (bruising, delayed wound healing; UBERON:0002097).
- **Tissue/cell level:** vascular **endothelium** (CL:0000115) and sub-endothelial matrix; connective tissue/ECM; **platelets** (α-granule PAI-1); hepatocytes and adipocytes as systemic PAI-1 sources; cardiomyocytes/cardiac interstitium in the fibrosis branch.
- **Subcellular (GO Cellular Component):** secreted/**extracellular space** (GO:0005615); PAI-1 is stored in platelet **secretory (α-)granules** (GO:0031093) and associates with extracellular matrix via vitronectin.
- **Localization / lateralization:** bleeding is site-dependent and typically focal to the site of trauma (not inherently lateralized); cardiac fibrosis tends to be regional (epicardial/posterior septum in models; patchy late gadolinium enhancement in humans).

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8. Temporal Development

- **Onset:** congenital (genotype present from birth); bleeding often first noted in childhood with trauma, circumcision, or menarche.
- **Onset pattern:** chronic/insidious trait with acute, provoked bleeding episodes.
- **Progression:** the hemostatic defect is **stable and non-progressive**; course is **episodic/provoked**. Characteristic *delayed* bleeding (initial clot then lysis) is a temporal signature.
- **Duration:** lifelong.
- **Patterns/critical periods:** menarche/menstruation, surgery/dental procedures, trauma, pregnancy/delivery, and the neonatal period (analogy from PAI-1-/- pups showing age-window vulnerability to intracerebral hemorrhage; Leroux et al. 2014,

P 24709679

) are windows of vulnerability and of intervention (peri-procedural antifibrinolytics). The cardiac-fibrosis branch is a slow, adult-emerging process.

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9. Inheritance and Population

- **Inheritance: autosomal recessive** (only homozygotes/compound heterozygotes are symptomatic; heterozygotes are asymptomatic for bleeding) (Fay 1997, P 9207454).
- **Penetrance/expressivity:** complete deficiency reliably produces a hyperfibrinolytic laboratory state; *clinical* bleeding is high-penetrance but **provocation-dependent** with variable expressivity (severity ranges from easy bruising to intracranial bleeds).
- **Anticipation / mosaicism:** not applicable / not reported.
- **Founder effect & consanguinity:** classic **founder effect** (c.699_700dupTA) in the **Old Order Amish**; endogamy increases homozygote frequency.
- **Carrier frequency:** low in the general population; elevated within the Amish founder community.
- **Epidemiology: extremely rare**; precise prevalence/incidence not established (only a handful of families worldwide plus the Amish kindred and small cohorts such as RBiN). Orphanet classifies it as a rare coagulation disorder.
- **Demographics:** no ethnic predilection beyond the Amish founder cluster; **sex ratio ≈ 1:1 genotypically**, but clinical burden is greater in women (menorrhagia, obstetric hemorrhage). Age distribution: lifelong, all ages.

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10. Diagnostics

Core principle: routine coagulation (PT, aPTT) and platelet function tests are **normal**, so a high index of suspicion plus specialized fibrinolysis testing is required (Jain & Acharya 2019, P 31427261).

- **Laboratory / biomarkers:**
- **PAI-1 antigen (ELISA):** undetectable in complete deficiency.
- **PAI-1 activity assay:** low/absent — but note the key pitfall that activity assays are poorly discriminating at the **low end** ("normal ranges begin at zero"), so a low value cannot by

itself prove deficiency (Mehta & Shapiro 2008,

P 19141166

).

- **Global fibrinolysis assays:** shortened **euglobulin clot lysis time**, shortened turbidimetric **clot lysis time**, and **enhanced plasmin generation** — functional confirmation of hyperfibrinolysis.
- Ancillary: tPA/α2-antiplasmin/plasminogen levels to exclude mimics.
- **Genetic testing (definitive): single-gene SERPINE1 sequencing** is the gold standard; bleeding-disorder/fibrinolysis **gene panels** or **WES/WGS** are useful when the phenotype is non-specific. CMA/karyotype/FISH/mtDNA/repeat testing are **not applicable**. Targeted variant testing (e.g., c.699_700dupTA) enables **cascade testing** in founder families.
- **Imaging:** cardiac **MRI with late gadolinium enhancement** detects the associated myocardial fibrosis (Khan 2021,

P 33439236

);
CT/MRI for suspected intracranial hemorrhage.
- **Clinical criteria / differential diagnosis:** no formal diagnostic criteria. Differentials among hyperfibrinolytic bleeding disorders: **α2-antiplasmin deficiency** (typically more severe bleeding), **Quebec platelet disorder** (uPA-rich platelets, gain-of-function), **tPA excess**, and acquired hyperfibrinolysis (DIC, liver disease, APL). Distinguish by specific factor assays and SERPINE1 genetics.
- **Screening:** no newborn screening. **Carrier/cascade screening** and prenatal/preimplantation testing are feasible in founder families via targeted variant analysis.

11. Outcome / Prognosis

- **Survival/life expectancy:** generally **normal** with trauma avoidance and antifibrinolytic management; mortality is driven only by severe unprovoked events (e.g., intracranial hemorrhage after head trauma). No disease-specific mortality rate is established.
- **Morbidity:** episodic bleeding, iron-deficiency anemia from menorrhagia, procedural/obstetric bleeding risk. Long-term, absolute deficiency is associated with **cardiac fibrosis** (potential arrhythmia/cardiomyopathy risk; Khan 2021,

P 33439236

).

- **Recovery:** bleeds resolve with antifibrinolytic therapy; the underlying trait is permanent.
- **Prognostic factors:** completeness of deficiency (absolute vs partial), access to antifibrinolytic prophylaxis, and exposure to hemostatic challenges. Reduced PAI-1 is paradoxically associated with favorable metabolic/aging markers (Khan 2017, P 29152572).
- **Prognostic biomarkers:** PAI-1 antigen/activity (deficiency confirmation); cardiac MRI late gadolinium enhancement (fibrosis burden).

12. Treatment

- **Pharmacotherapy — first line: antifibrinolytic lysine analogues**: **tranexamic acid** (NCIT:C835; CHEBI:48669) and **ε-aminocaproic acid** (NCIT:C28987; CHEBI:28402). They competitively block plasminogen's lysine-binding to fibrin, stabilizing clots, and are effective both to **treat** and **prevent (peri-procedural prophylaxis)** bleeding (Fay 1997, P 9207454; Jain & Acharya 2019, P 31427261). Documented peri-operative regimens for dental/oral surgery (Morimoto 2004, P 15357795; Kajihara 2003, P 12632627; Tanimura 1994, P 8015955).
- **Pharmacogenomics:** none specific.
- **Advanced/experimental therapeutics: PAI-1 mRNA lipid-nanoparticle replacement** restores circulating PAI-1 in preclinical models (Ferraresso et al. 2025, P 40917690) — a mechanism-directed future therapy. No approved gene therapy, cell therapy, siRNA/

ASO, or monoclonal antibody for this indication. Recombinant/plasma-derived PAI-1 is not a standard product.

- **Supportive/interventional:** local hemostatic measures, iron supplementation/transfusion for anemia, hormonal management of menorrhagia (combined with tranexamic acid), and careful peri-operative/obstetric planning.
- **Treatment outcomes:** most bleeding events are controlled with antifibrinolytics (Mehta & Shapiro 2008,

P 19141166

). Adverse effects of antifibrinolytics: GI upset, and theoretical thrombosis risk (low in this deficiency context).
- **Strategy/personalization:** genotype-confirmed deficiency → lifelong plan of **antifibrinolytic prophylaxis around hemostatic challenges**, trauma precautions, and (where relevant) cardiac surveillance. NCIT intervention terms: Tranexamic Acid (C835), Aminocaproic Acid (C28987), Antifibrinolytic Agent (C77096).

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13. Prevention

- **Primary prevention:** not preventable (germline). **Preconception genetic counseling** and **carrier/cascade screening** in founder families reduce recurrence; prenatal/preimplantation genetic testing available.
- **Secondary prevention:** early molecular diagnosis in bleeders with normal routine coagulation tests; identification of at-risk relatives.
- **Tertiary prevention (complication avoidance): peri-procedural and menstrual antifibrinolytic prophylaxis**, trauma avoidance/contact-sport counseling, iron repletion, obstetric planning, and cardiac monitoring for fibrosis.
- **Immunization / public-health / environmental measures:** not applicable.
- **Counseling:** autosomal-recessive recurrence risk (25% for carrier×carrier couples); explain asymptomatic carrier status and provocation-dependent bleeding.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** human **SERPINE1** (Gene 5054) is conserved; mouse **Serpine1** (Gene 18787; *Mus musculus*, **NCBI Taxon 10090**), rat Serpine1 (Rattus norvegicus, Taxon 10116). Orthologs exist across vertebrates.
- **Natural disease in animals:** no well-established spontaneous congenital PAI-1-deficiency disease is cataloged in companion animals (not a recognized OMIA entry); knowledge derives from engineered models rather than natural disease.
- **Comparative biology:** mechanism (serpin inhibition of tPA/uPA → fibrinolysis control) is evolutionarily conserved. A key **species difference**: mouse knockouts have mild hyperfibrinolysis and thrombosis resistance **without impaired baseline hemostasis**, whereas humans show provoked/delayed bleeding (Lijnen 1996,

P 8739849

).
- **Transmission / zoonosis:** not applicable.

15. Model Organisms

- **Mouse (primary model):** constitutive **Serpine1-knockout** mice are viable and fertile; they display a **mild hyperfibrinolytic state and resistance to venous/arterial thrombosis without impaired hemostasis** (Lijnen 1996,

P 8739849

;
Wu & Zhao 2002,

P 12769655

).
Disease-relevant bleeding phenotypes are **challenge-dependent**:
- Neonatal (postnatal day 3–5) PAI-1-/- pups develop **intracerebral hemorrhage** after minor trauma, modeling age-windowed vascular vulnerability (Leroux et al. 2014,

P 24709679

).
- Angiotensin II or norepinephrine infusion causes **epicardial hemorrhage then posterior-septal cardiac fibrosis** in PAI-1-/- hearts (Pettey et al. 2025,

/ P 41332700 42946663), recapitulating the human cardiac-fibrosis association.

- **In vitro / cellular models:** CRISPR/Cas9 **PAI-1-knockout human endothelial cells** (markedly increased plasmin generation; disordered migration, reversed by aprotinin/recombinant PAI-1/EACA; Sano et al. 2026,) P 42946217 and patient-derived **iPSC-cardiomyocytes** (Khan 2021,). P 33439236
- **Genetic model types available:** constitutive knockout; transgenic PAI-1-overexpressing mice (opposite, pro-thrombotic phenotype) as comparators; cellular CRISPR knockouts and iPSC lines.
- **Phenotype recapitulation & limitations:** models capture the **hyperfibrinolysis/anti-thrombotic axis** and, under stress, the **hemorrhage→fibrosis** branch; the main limitation is that mice **do not spontaneously bleed at baseline**, underscoring species differences in hemostatic reserve.
- **Resources:** MGI (Serpine1), IMPC/IMSR for knockout strains.

Key Supported vs. Refuted Hypotheses

Supported: - SERPINE1 biallelic null variants cause complete PAI-1 deficiency (frameshift c.699_700dupTA) → autosomal-recessive bleeding (Fay 1997; Khan 2021). - Mechanism is hyperfibrinolysis from unopposed tPA/uPA → excess plasmin → premature clot lysis (Jain & Acharya 2019; Sano 2026). - Antifibrinolytics are effective first-line therapy (Fay 1997; multiple case reports). - Chronic absolute deficiency is associated with cardiac fibrosis (Khan 2021; mouse models). - PAI-1 loss has protective pleiotropy in senescence/metabolism/longevity (Khan 2017).

Refuted / not supported: - Heterozygous carriers do **not** have a clinically significant bleeding disorder (Fay 1997). - Spontaneous (unprovoked) major bleeding is **not** typical, unlike in procoagulant factor deficiencies (Mehta & Shapiro 2008). - Mouse baseline hemostasis is **not** impaired by PAI-1 loss (contrasts with humans) (Lijnen 1996).

Limitations and Future Directions

- **Rarity** limits prevalence estimates, genotype–phenotype breadth, and controlled treatment data; most evidence is from one founder kindred and small case series.
- **Assay limitations** (PAI-1 activity insensitive at low range) make molecular confirmation essential.
- Open questions: natural history and management of the **cardiac-fibrosis** complication; clinical translation of **PAI-1 mRNA-LNP replacement**; and whether partial/quantitative deficiencies cause milder, under-recognized bleeding.

Evidence types: human clinical (Fay 1997; Khan 2017/2021; Mehta & Shapiro 2008; case reports), reviews (Jain & Acharya 2019; Mehta & Shapiro 2008), model organism (Lijnen 1996; Leroux 2014; Pettey 2025; Wu & Zhao 2002), in vitro (Sano 2026; Ferraresso 2025), structural/computational (Sharp 1999; Kihn 2022).
